

MN572: Reliability Engineering (Outline)

- **Basic Concepts of Reliability**
- **Component Reliability and Hazard Models**
- **Modelling of System Reliability**
- **Design for Reliability**
- **Equipment Reliability Analysis**

Objective of the course

- ✓ An elective course designed to apply engineering knowledge and specialist techniques to prevent or to reduce the likelihood of failures.
- ✓ To understand the concepts of Reliability, Maintainability and Availability.
- ✓ It is designed to identify and correct the causes of failures that do occur despite the efforts to prevent them.
- ✓ Another aim of this course is to apply methods for estimating the likely reliability of new designs, and for analyzing reliability data.

Outcome of the course

- ✓ Have a working knowledge of the techniques of reliability estimation and analysis.
- ✓ Learn to analyze failure, failure modes and causes of failures.
- ✓ Ability to apply learned concepts to improving the maintenance, the maintainability, hazard risk and the reliability of a system.
- ✓ Be able to carry out the Failure Mode Effect Analysis (FMEA) and Failure Mode Effect Criticality Analysis (FMECA) and Fault Tree Analysis (FTA)

MN572: Reliability Engineering (Outline)

Basic Concepts of Reliability

Introduction to Reliability Engineering. Failures and failure modes. Causes and mechanism of failures. Maintainability and availability.

Component Reliability and Hazard Models

Failure distributions. Constant and Time dependent hazard models. Exponential, Normal, Lognormal and Weibull failure models.

Modelling of System Reliability

Reliability block diagrams. Series and parallel configurations, Redundancies. Markov Chain.

Equipment Reliability Analysis

Type of failure data. Sources of failure data. Classification of failure data. Identification of Failure distributions. Goodness-of-fit Test. Analysis of equipment reliability.

Design for Reliability

Reliability allocation. Designing for higher reliability. Failure analysis: FMEA, FMECA, FTA. Reliability optimization.

List of Books

- 1) Ebeling, C. E. (2004), **An Introduction to Reliability and Maintainability Engineering**, Third Edition, Tata McGraw-Hill Publishing Company Limited
- 2) O'Connor, D.T. (2002), **Practical Reliability Engineering**, 4th edition, Wiley
- 3) Ernest J. Henley (1991), **Probabilistic Risk Assessment: Reliability Engineering, Design, and Analysis**, IEEE Press.
- 4) Birolini, A. (2006), **Reliability Engineering: Theory and Practice**, 4th edition, Springer
- 5) Smith, David J. (2002), **Reliability, Maintainability and Risk**, 6th edition, Butterworth-Heinemann Limited
- 6) Pham, Hoang (2006), **Handbook of Reliability Engineering**, Springer
- 7) Gupta, Suprakash and Bhattacharya, Jayanta (2011), **Aspects of Reliability and Maintainability in Bulk Material Handling System Design and Factors of Performance Measure**, In **Design and selection of bulk Material Handling Equipment and Systems Vol. – I**, edited by Bhattacharya, Jayanta, Wide Publishing, Kolkata, pp 153-188.

Course Requirements

- **Attendance Requirement 75%, should be self monitored.**
- **Total Marks 100.**

Sl. No.	Exam/ Work	Total Weightage (%)
1	Mid-Term Exam	30
2	End-Sem Exam	50
3	Quiz/Assignment/Test/term paper	20
4	Total	100

Purpose of reliability Engineering

To develop methods & tools to evaluate and demonstrate reliability, availability, maintainability, and safety (RAMS) of components, equipment and systems, as well as to support development and production engineers in building in these characteristics.

The objectives of reliability engineering, in the order of priority are:

- ✓ To apply engineering knowledge and specialist technique to prevent or to reduce the likelihood or frequency of failures.
- ✓ To identify and correct the causes of failures that do occur despite the efforts to prevent them.
- ✓ To determine ways of coping with failures that do occur, if their causes have not been corrected.
- To apply methods for estimating the likely reliability of new designs, and for analyzing reliability data.

Some examples of failure

- ✓ Lawn-mower casing cracks – Aluminum construction have insufficient strength to withstand the stress.
- ✓ Washing machine fails – motor wear out after a normal life
- ✓ Car battery go dead – completed normal life and drained off.
- ✓ A toaster oven electrical plug burn – poor design
- ✓ A water heater leak – partly due to lack of preventive maintenance
- ✓ TV-remote control quit functioning – random electronic component failure
- ✓ A stereo amplifier quit functioning – failure caused by an open at solder joint
- ✓ An automobile engine starter fails – wear out after a normal life.
- ✓ House roof leak – poor construction

Consequences of failure

- ✓ Some failures cause – inconvenience + economic impact
- ✓ Other causes – personal safety problem even injury or loss of life.

Effects of failure are:

- ✓ economic
- ✓ safety

Impact of product and system failures varies from minor inconvenience and costs to personal injury, significant economic losses and death.

Failures

A failure occurs when an item stops performing its required function. It occurs due to permanent damage to the system. System starts functioning only after repair/replacement. Failures are classified according to the **mode, cause, effect** and **mechanism**.

MODE: The mode of failure is the symptom (local effect) by which a failure is observed; i.e., the way something might fail; for example, opens, shorts, creep, cracking, seizes or fatigue for mechanical components.

Failures cont.....

CAUSE: Failure causes are defects in design, process, quality, or part application, which initiate a process which leads to failure. The cause of a failure can be intrinsic, due to weakness (macro defects caused by manufacturing flaws) in the item and/or wear out, or extrinsic, due to errors, misuse or mishandling during the design, production, or use. Extrinsic failures are caused due to transient overload --- electrical, thermal, mechanical.

EFFECT: The effect (consequence) of a failure can be different if considered on the item itself or at higher level. A usual classification is non-relevant, partial, complete, and critical failures. Since a failure can also cause further failures in an item, a distinction between primary failure and secondary failure is important.

MECHANISM: Failure mechanism is the physical, chemical, or other processes resulting in a failure, e.g., excessive deflection, buckling, ductile fracture, brittle fracture, impact, creep, relaxation, thermal shock, wear, corrosion, stress corrosion cracking, and various types of fatigue, etc.

Catastrophic and Degradation Failures

- Failures are also classified as sudden and gradual. Sudden and complete failures are termed cataleptic (or catastrophic) failures, gradual and partial failures are termed as degradation failures.
- Catastrophic failures are complete & sudden, like a light bulb failing. Degradation is gradual or partial like an electronic unit moving outside specification. In case of an electrical supply, a complete loss of power would be a catastrophic failure, while voltage or frequency deviation would be regarded as degradation.

Primary and Secondary Failures

- A failure is primary if it is **not caused** by a failure in another part of the system, and secondary if it is the result of the failure of another part of the system. As an example, the normal failure of a light bulb would be a primary failure, but failure following a severe overvoltage would be a secondary failure because it is the result of a failure elsewhere.
Reliability data can only give information about primary failures.

As failure is not the only cause for an item being down, the general term used to define the down state of an item (not caused by a preventive maintenance, other planned actions, or lack of external resources) is **fault**. **Fault is thus a state of an item and can be due to a defect or a failure.**

Causes of Failures

- Bad engineering design – inherently incapable design.
- Faulty construction or manufacturing process – house roof leaking
- Human error
- Poor maintenance – A water heater leak
- Inadequate testing and inspection
- Improper use
- Lack of protection against excessive environmental stress

Reliability Improvement

- Manufacturer can be held liable for failing to account properly for product safety and reliability
- Engineers are responsible for product design must therefore include both reliability and maintainability as design criteria.
- Reliability and maintainability engineering attempts to study, characterize, measure and analyze the failure and repair of systems to improve their operational use by increasing their design life, eliminating or reducing the likelihood of failures and safety risks, and reducing downtime, thereby increasing available operating time.
- Reliability and maintainability are an important part of the engineering design process and a necessary functions in life – cycle – costing (LCC), cost benefit analysis, operational capability studies, repair and facility resourcing, inventory and spare parts requirements determinations, replacement decisions, and the establishment of preventive maintenances programs.

Reasons for Improving Reliability

The justification for a reliability improvement program usually falls into one or more of the following five categories.

Increased production: every extra hour of operation leads to increased revenues, which are often profitable because all the fixed and semi-fixed costs have already been covered.

Increased productivity: plant staff have more time to focus on revenue-generating projects rather than coping with crises.

Reduced maintenance expense: save money by needing less repair work.

Improved safety: reliability improvements lead to improvements in safety.

Reduced environmental problems: an unplanned shutdown often results in large, unexpected discharges of process materials into the environment.

Reliability Improvement cont....

A product has utility because it satisfies a need. Its utility value depends on:

- ✓ Versatility
- ✓ Ease of use
- ✓ Safety
- ✓ Aesthetics
- ✓ Reliability

The primary reason for reliability and maintainability Engineering is to improve the reliability and availability of the product or system being developed and thereby add to its value (utility).

Reliability Improvement cont....

Ways of improving reliability during design

- ✓ For critical & high failure rate components- redundancy,
- ✓ Designing excessive strength into components and careful selection of material- high FOS
- ✓ Derating – Operating the system below its rated stress level.
- ✓ Choice of technology – mechanical vs electrical/electronic, transistors vs IC
- ✓ Reducing complexity – reduce no of components.

Availability Improvement

If the number of components increases, to achieve a target reliability, each component must have a high reliability. Once the limits of reliability improvement have been reached, further gains in product availability may be obtained by decreasing downtime through good maintainability design.

Good quality control during manufacturing will maintain the reliability of the product design. Inspection and acceptance sampling help to ensure raw materials and supplier parts are of standard quality.

Once the product is operational, failure may be reduced through

- ✓ Preventive maintenance (PM)
- ✓ Sound parts replacement policy
- ✓ Engineering modification
- ✓ Careful attention to environmental conditions and operating loads.

$$\frac{\text{Mean Time to Failure (MTTF)}}{\text{MTTF} + \text{Mean Time to Repair (MTTR)}}$$

Availability Improvement

Once a system is put into operation its downtime can be reduced by increasing its maintainability with proper amount of repair resources like:

- Maintenance technicians
- Test equipment
- Available spare parts

Secondarily,

- ✓ Skill levels
- ✓ Resupply lead time
- ✓ Maintenance training
- ✓ Ease of use of technical manuals

Concept of reliability

Reliability is the **trustworthiness** and close to **consistency** in performance. In statistics, it is inversely related to random failure and is often expressed as a probability. Reliability is a characteristic of an item. From a qualitative perspective, it is an item's ability to remain functional. Quantitatively, it specifies the probability that **no operational interruption** will occur during a stated time interval. This does not mean that redundant parts cannot fail; such parts can fail and be repaired in time. The concept of reliability thus applies to both non-repairable and repairable items. To make sense, a numerical statement of reliability (e.g., $R = 0.9$) must be accompanied by the definition of the required function, the operating conditions, and the mission duration.

Definition of reliability

Reliability is the bedrock of any dependable system. Reliability goes beyond just uptime – it's the measure of a system's ability to perform its intended function consistently without unexpected failures. It is like a trustworthy friend whom one can always count on. A reliable system not only operates smoothly but also **builds confidence** in its users.

- In engineering, **Reliability** is usually defined as *“the probability that a system or component will perform its intended functions under stated operation conditions for **a given period of time**”*.

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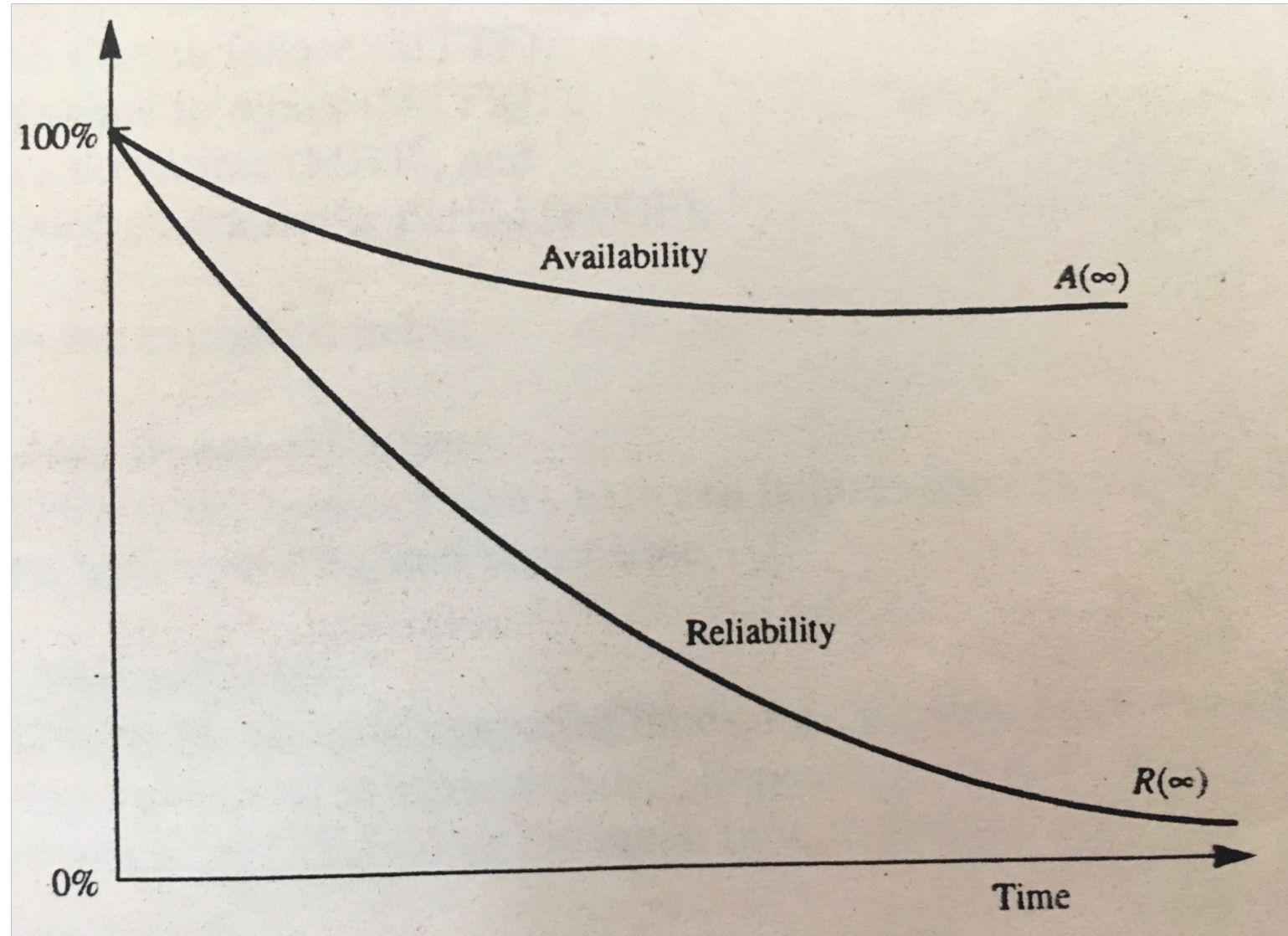
Availability

Availability is like the dynamic sibling of reliability, closely intertwined yet with its unique characteristics. While reliability focuses on the consistency of performance, availability zooms in on how often a system is operational and ready to deliver its services. If reliability is the friend you can trust, availability is the friend who is always ready for an adventure. These two concepts are inseparable – a reliably performing system naturally contributes to high availability.

Reliability is the probability that a fault will not occur during a specified time interval, whereas availability is the probability that a fault will not exist at a specified time interval.

Definition of availability

Availability is defined as “the probability that a repairable component or system is performing its required function at a given point in time when used under stated operation conditions”.



Maintainability: Keeping the System Fit

Maintainability is the measure of how quickly and efficiently a system can be restored to operational state/status after a failure or during planned maintenance. Maintainability complement reliability by focusing on the downtime aspect of the system. Even the most reliable systems may encounter failures or require maintenance. This is where maintainability steps in, minimizing downtime and ensuring the system can swiftly recover from setbacks.

Definition of maintainability

- **Maintainability** is defined to be *“the probability that a failed component or system will be restored or repaired to a specified condition within a stated period of time when maintenance is performed in accordance with prescribed procedures”*.

Most calculations involving maintainability assume that the item is returned to a condition that is **“as good as new”**. In fact, the assumption is rarely true; most items are restored to a condition that is somewhere between **“as good as new”** and **“as good as old”**, i. e., worse than new, but better than it was at the time of failure.

Reliability versus Quality

Reliability is closely associated with product quality and is often considered a **subset** of it. Quality can be defined qualitatively as the extent to which the product satisfies users' (customers') requirements. Product quality is a function of design and conformance to design specifications. It also depends on the production system and on adherence to manufacturing procedures and tolerances. Quality is achieved through a good quality assurance program. Quality assurance is a planned set of processes and procedures necessary to achieve high product quality.

On the other hand, reliability concerns how long the product continues to function once operational. A poor-quality product is likely to have poor reliability, and vice versa. Reliability may be viewed as the quality of a product's operational performance over time; as such, it extends quality into the time domain.